Data Flow Diagram (DFD)

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Team ID	Rahul-DS633
Project Name	CrimeVision AI - Multimodal Crime Scene Reconstruction
Maximum Marks	5 Marks

1. Level 0: Context Data Flow Diagram

The Level 0 Context Diagram represents CrimeVision AI as a single central entity interacting with external entities (Forensic Investigator, Evidence Storage, Groq Cloud Reasoning, and Judicial Court System).

External Entity	Data Inflow to CrimeVision AI	Data Outflow from CrimeVision AI
Forensic Investigator	• Crime Scene Photos (JPG/PNG) • 911 Audio / Witness Tapes (WAV/MP3) • CCTV Surveillance Video (MP4) • Natural Language Queries	• Interactive 3D Spatial Crime Scene • Second-by-Second Synced Timeline • AI Chat Answers & Forensic Summaries • Court-Admissible PDF Reports
Model Registry & Storage	• Model weights (RT-DETR, CLIP, Whisper) • Historical case sessions JSON	• Persisted Session context & timeline records • Generated Forensic PDF dossiers
Groq Cloud Reasoning API	• Groq LLaMA-3.3 forensic hypotheses • Contextual answers & multilingual translations	• Structured JSON prompts containing detected objects, audio facts, and temporal keypoints

2. Level 1: Functional Subsystem Data Flow Decomposition

Process ID & Name	Input Data	Internal Algorithm / Engine	Output Data & Store
P1: Vision Pipeline	2D Scene Image File	RT-DETR object detection + Depth estimation + 3D coordinate projection.	BBoxes, classes, 3D Mesh [X,Y,Z] -> Session Store
P2: Audio Pipeline	Audio waveform (.wav)	Whisper ASR Speech-to-Text + Acoustic event detection + Groq translation.	Transcript, crime type, 4-language translations -> Session Store
P3: Video Pipeline	CCTV Footage (.mp4)	1.0 Hz Frame extraction + BLIP captioning + Suspicious activity classifier.	Chronological second-by-second timeline -> Session Store
P4: Multimodal Fusion	Outputs from P1, P2, P3	Cross-modal timestamp alignment & fact consistency verification.	Unified Case Dossier & Verified Facts -> Session Store
P5: PDF Generation	Session context dictionary	ReportLab Platypus document synthesis with XML character sanitation.	Forensic_Report_{id}.pdf -> Reports Storage